Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/31/2006
PI Name:	Paloski, William H Ph.D.		
Project Title:	Age Associated Differences in Postural Equilibrium Control: Implications for Space Flight		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:	National Institutes of Health/National Institute on Aging	**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	whpaloski@uh.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	(713) 743-9272
Organization Name:	University of Houston		
PI Address 1:	Health & Human Performance		
PI Address 2:	3855 Holman St, Garrison 104		
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77204-6015	**Congressional District:**	18
Comments:	PI moved to University of Houston in September 2008, from NASA JSC.		
Project Type:	Ground	**Solicitation / Funding Source:**	OTHER
Start Date:	10/01/1999	**End Date:**	03/31/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	N/A		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	National Institutes		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Aging is associated with greater susceptibility to physical disability and frailty. Poor balance is an important contributor for falls in community dwelling adults [Tinetti, 1988]. Elderly individuals, both women and men, show increasing body sway, less secure base of support, greater dependence on sensory cues from vision [Sundermier, 1996], as vestibular and somatosensory systems decline in performance. There may be some gender differences in the nature of the aging changes [Simoneau, 1992; Wolfson, 1994]. In addition to changing sensory systems, motor control declines with age, with slower response times, accuracy of movement and losses of muscle strength [Teasdale, 1991]. For this investigation, we propose that aging leads to (1) changes in vestibular, visual and sensory systems that will mimic the loss of otolith stimulation with increased emphasis on visual spatial information; (2) reduced effectiveness of both closed and open loop mechanisms; and (3) is similar to responses observed in astronauts following extended exposure to		

	microgravity. We plan to perform dynamic posturography on participants in the Baltimore Longitudinal Study of Aging using the same protocol as being used by NASA in testing their astronauts. The methods allow the assessment of the sensory systems primarily responsible for postural equilibrium from quantitative analysis of the effectiveness of the strategies used to maintain upright stance under conditions of normal, absent, and altered visual and/or ankle proprioceptive sensory stimuli. The data will be compared to the astronaut data to examine the implications of age on potential long term space flight in middle aged and elderly individuals.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results of this research can be used by the NASA and National Institutes of Health (NIH) communities to better understand the recovery process of inner ear patients and to improve rehabilitation treatments on Earth. By characterizing the etiologies of balance dysfunction associated with aging, this information will also be used to design techniques for restoring lost movement and balance control capabilities in astronauts.
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/09/2021)